

A computable asymptotically decreasing sequence with no computable full-tail rate

A candidate full answer to Powell's footnote conjecture

Verdict. **Candidate full solution, likely valid.** There is a uniformly computable sequence of nonnegative rational numbers that is asymptotically decreasing but admits no computable function $\phi(\varepsilon, N)$ returning a witness for Powell's full-tail condition (5).

Source. Thomas Powell, "A new metastable convergence criterion and an application in the theory of uniformly convex Banach spaces," arXiv:1902.09284v2 (2019), Section 2, PDF pages 4–5 [1].

Answer in one line. At the computably scheduled stage at which machine e is observed to halt, place a spike of height 2^{-e} ; put zero at every other stage. The spikes converge to zero, but a computable full-tail rate would reveal a stage after which the e th spike cannot occur and would therefore decide the halting problem.

The exact source question

For a nonnegative sequence (x_n) , Powell calls

$$\forall \varepsilon > 0 \forall N \exists n \forall i \geq n (x_i \leq x_N + \varepsilon) \quad (\text{AD})$$

asymptotic decreasingness. Theorem 2.2 of the source proves that this is equivalent to convergence of (x_n) to its infimum. Immediately after the theorem, the source gives an example with a noncomputable first term and asks in footnote 2 whether the sequence can instead be chosen computable.

and by (2) some n such that $x_i \leq x_N + \varepsilon \leq a + 2\varepsilon = l - 2\varepsilon$ for all $i \geq n$, contradicting $x_n \rightarrow l$. Therefore $l = d$. $\square$

In general, there are sequences of nonnegative reals which have no computable metastable rate of asymptotic decreasingness, by which we mean a procedure $\phi(\varepsilon, N)$ which returns some n satisfying (5). A trivial example would be to take any decreasing Specker sequence $(a_n)_{n \in \mathbb{N}}$ which converges to some noncomputable infimum d , and define $(x_n)_{n \in \mathbb{N}}$ by $x_0 := d$ and $x_{n+1} := a_n$. Then

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$\phi(\varepsilon, 0)$ would be a computable rate of convergence for the Specker sequence, which is impossible by definition (2).

Thus we are interested in the metastable version of asymptotic decreasingness, namely the reformulation of (5) analogous to metastability in the usual

²A. Sipos asks if we could strengthen this by choosing $(x_i)_{i \in \mathbb{N}}$ to be *computable*, since in our example, $x_0 = d$ is a noncomputable real. We conjecture that this is indeed possible, but a more detailed discussion at this point would take us too far afield.

The source describes a procedure $\phi(\varepsilon, N)$ that returns an n satisfying the *full tail* in (AD). We call this a *tail rate* below to distinguish it from the finite-window functional $\Gamma(\varepsilon, g, N)$ introduced just after the footnote.

Proof intuition

A computable sequence can converge to the computable limit zero without any computable convergence modulus. The standard obstruction is to reserve one visible spike for each Turing machine. Machine e receives height 2^{-e} , so only finitely many machines can ever create spikes above a fixed tolerance. Consequently the sequence converges to zero. But a tail bound below $2^{-(e+1)}$ certifies that the e th spike will never appear; this certificate decides whether machine e halts.

The extra zero term $x_0 = 0$ turns that convergence modulus directly into the source's requested witness $\phi(\varepsilon, 0)$, while every term remains a computable rational.

Formal construction and theorem

Fix an effective enumeration $(M_e)_{e \in \mathbb{N}}$ of Turing machines on blank input. If M_e halts after exactly h_e computation steps, schedule its unique spike at

$$s_e = e + h_e + 1.$$

For $n \geq 1$, define

$$x_n = \max(\{2^{-e} : s_e = n\} \cup \{0\}), \quad x_0 = 0. \quad (1)$$

For fixed n , only $e < n$ can contribute, and whether M_e halts after exactly $n - e - 1$ steps is decidable by a finite simulation. Thus (1) is a uniformly computable sequence of rational numbers.

Theorem 1. *The sequence $(x_n)_{n \in \mathbb{N}}$ in (1) is a computable sequence in $[0, 1]$, converges to its infimum 0, and hence is asymptotically decreasing. There is no computable function ϕ on positive rational ε and $N \in \mathbb{N}$ such that*

$$\forall i \geq \phi(\varepsilon, N), \quad x_i \leq x_N + \varepsilon. \quad (2)$$

In particular, the sequence gives the computable strengthening conjectured in footnote 2 of the source.

Proof. Computability and the bound $0 \leq x_n \leq 1$ follow from the finite simulation described above. To prove convergence, fix $\varepsilon > 0$ and choose E with $2^{-E} \leq \varepsilon$. Only the finitely many machines $M_0, \dots, M_{E-1}$ can generate a spike larger than ε , and each generates at most one spike. After the last such spike (among those machines that halt), every x_n is at most ε . Hence $x_n \rightarrow 0$. Since $x_0 = 0$ and all terms are nonnegative, $\inf_n x_n = 0$. Therefore (x_n) is asymptotically decreasing, either directly from (AD) or by Powell's Theorem 2.2.

Suppose, for contradiction, that a computable ϕ satisfies (2). Given an index e , compute

$$n_e = \phi(2^{-(e+1)}, 0).$$

Simulate M_e for n_e steps. If it halts, answer yes. If it has not halted, it can never halt later: a later halt after $h_e > n_e$ steps would put the spike $x_{s_e} = 2^{-e}$ at the stage $s_e = e + h_e + 1 > n_e$, contradicting (2), because $x_0 + 2^{-(e+1)} = 2^{-(e+1)} < 2^{-e}$. Thus this algorithm decides the halting problem, contradicting its undecidability [2]. No such computable ϕ exists. $\square$

Verification and scope

- Every x_n is an exactly computable dyadic rational; the construction does not hide a noncomputable real in any term.
- The convergence argument is classical rather than effective: locating the last large spike would decide finitely many halting questions. This is precisely why a computable tail rate cannot exist.
- The conclusion concerns the full-tail witness $\phi(\varepsilon, N)$ described before footnote 2. It does *not* assert that the later finite-window functional $\Gamma(\varepsilon, g, N)$ in equation (6) is noncomputable. For a fixed computable g , finite candidate blocks for this particular rational sequence can in fact be searched effectively.
- The result answers only the computability footnote. It does not alter the Banach-space fixed-point estimates developed later in the source.

Novelty check and review recommendation

On 2026-07-19 we searched the exact conjectural sentence, the phrases “computable metastable rate of asymptotic decreasingness” and “computable asymptotically decreasing sequence,” the source title and author, and close halting-problem variants. We found the source and later works citing its quantitative fixed-point results, but no later answer to this footnote. The spike mechanism is standard computability folklore, so novelty confidence is conservative even though the exact application was not found.

Human review recommendation. Confirm that “computable sequence” is intended in the standard uniform termwise sense used here, and that the footnote asks about the full-tail ϕ rather than equation (6)’s finite-window Γ . Under those conventions, the proof is complete.

References

- [1] T. Powell, *A new metastable convergence criterion and an application in the theory of uniformly convex Banach spaces*, J. Math. Anal. Appl. 478 (2019), 790–805; [arXiv:1902.09284](#).
- [2] A. M. Turing, *On computable numbers, with an application to the Entscheidungsproblem*, Proc. London Math. Soc. (2) 42 (1937), 230–265.